Dataset attributes & mathematical model

Review edition 1.0 | 26 October 2026 | ReCell research prototype | India C&I dashboard context

Purpose. Document the prominent dataset attributes actually selected for training and formalize the equations already implemented in ReCell. This is an implementation review, not a claim that the model has been validated on Indian industrial packs.

The project has two mathematical problems

Stage	Input and output	Objective
Health forecasting	Past battery checkpoint features → future SOH at the next two checkpoints	Learn point and quantile forecasts that generalize to a held-out cell.
Storage dispatch	Held-out health forecasts + hourly solar/load + INR assumptions → charging schedule	Minimize energy charges plus weighted estimated battery wear.

Six selected training attributes

1. Past state of health (SOH)
2. Elapsed days since first observed checkpoint
3. Cumulative charge-plus-discharge throughput (Ah)
4. Pulse-derived resistance (ohms)
5. Commercial duty-cycle indicator
6. Documented first-life-end capacity (Ah)

Selection is not importance. These are the configured predictors. No causal ranking or claim that one predictor dominates the others is made in this document.

Evidence and scope

The saved experiment contains **111 checkpoints from 6 cells**, six forecasting methods, three seeds and two forecast horizons. The horizon is measured in diagnostic checkpoints, not fixed days. The dashboard uses a 720-hour German household solar/load proxy with newly optimized illustrative INR economics.

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Read the assumptions alongside every result: the wear proxy is linear, health is fixed during replay, solar/load are known in advance, and site-specific Indian C&I economics remain to be measured.

Training attribute dictionary

Feature order is taken from the saved LOBO experiment configuration, not inferred from column names.

Configured attribute	Unit	Definition and role	Observed range
soh	Ratio	Measured diagnostic discharge capacity divided by the same cell's first observed second-life capacity. Past SOH is a predictor; future SOH is the label.	0.950494 to 1.000000
elapsed_days	Days	Checkpoint end timestamp minus the cell's first checkpoint end timestamp; seconds divided by 86,400.	0-806.736
cumulative_ah	Ah	Sum of charge Ah and discharge Ah from cycling sessions ending before the diagnostic test begins.	65.129-1,422.087
pulse_resistance_ohm	Ohms	Mean accepted pulse resistance, ΔV/ΔI, from the diagnostic test near the reviewed 3.63 V condition.	0.031889 to 0.038869
duty_commercial	Binary	Static reviewed duty metadata: 1 = commercial, 0 = residential. This describes the cell aging protocol, not Indian site classification.	0 or 1
first_life_capacity_ah	Ah	Documented first-life-end capacity from the paper's Table 2, mapped by cell identity; not fitted from future second-life measurements.	4.4568-4.7055

All six selected columns have **zero missing values in the current 111-row checkpoint table**. This does not mean the raw dataset has no missing data. Median imputation is still fitted inside each training fold.

Raw measurements are not the same as model features

Current, test time, voltage, discharge-capacity counters, step identifiers and timestamps support capacity extraction and quality checks. Charge/discharge counters support exposure summaries. The current forecaster receives the six derived/static attributes above, not raw current or voltage traces.

Not selected as direct predictors: temperature, EIS spectra, cumulative Wh, cycling-session count, raw voltage/current traces and cell ID. Some are retained for auditing or analysis. Temperature availability and sensor interpretation require further review.

Three capacity references: first-life-end capacity is a predictor; first observed second-life capacity normalizes SOH; 5 Ah nominal cell capacity converts SOH into a dispatch health fraction. These quantities must not be interchanged.

From measurements to samples

Indices: i = battery cell; k = diagnostic checkpoint; h = forecast horizon; j = feature; t = hourly dispatch interval.

1 / Extract discharge capacity and construct SOH

$$Q_{i,k} = -\frac{1}{3600}\sum_{s=1}^{4}\sum_{n=1}^{N_s-1}\frac{I_{s,n}+I_{s,n+1}}{2}(\tau_{s,n+1}-\tau_{s,n})$$

Q is in Ah, current I is in A and test time τ is in seconds. Discharge current is negative in these exports, hence the leading minus sign. Integrate four accepted low-rate discharge segments; exclude rests, charging and short pulses.

Extraction checks include median discharge current magnitude 0.15-0.30 A, segment duration at least 300 s, sample gaps at most 5 s, the reviewed voltage span, a 1% counter-agreement tolerance and capacity plausibility bounds 2.0-5.5 Ah.

$$y_{i,k} = \frac{Q_{i,k}}{Q_{i,0}}, \qquad T_{i,k} = \frac{\text{timestamp}_{i,k} - \text{timestamp}_{i,0}}{86400}$$

G1 example: first observed capacity 4.5767 Ah; last observed capacity 4.4028 Ah. Its last SOH is 0.96200, or 96.20% of its second-life starting reference, not factory-new capacity.

2 / Build causal exposure and the feature vector

$$A_{i,k} = \sum_{s\,:\,\text{end}_s \le \text{startRPT}_{i,k}} \left(A_s^{\text{charge}} + A_s^{\text{discharge}}\right)$$

$$\mathbf{x}_{i,k} = [y_{i,k},\ T_{i,k},\ A_{i,k},\ R_{i,k},\ d_i,\ Q_i^{\text{FL}}]^{\mathsf{T}}$$

Counter resets are handled using positive increments and observed reset values; the unknown initial counter offset is omitted. The pulse resistance uses R = ΔV/ΔI and averages only accepted diagnostic pulses.

3 / Fit preprocessing only on training cells

$$\widetilde{x}_j = \frac{\text{impute}_{m_j}(x_j) - \mu_j}{\sigma_j}$$

For each fold, median m, mean μ and scale σ are fitted on unique training-cell checkpoints. Validation/test values never fit these statistics. A constant training column uses the scaler's unit scale.

$$X_{i,k} = [\widetilde{\mathbf{x}}_{i,k-L+1}, \ldots, \widetilde{\mathbf{x}}_{i,k}], \quad \mathbf{y}^{+}_{i,k} = [y_{i,k+1}, y_{i,k+2}]$$

Origins require at least three past checkpoints before history truncation. Limited-history tests retain L = ceil[f(k+1)] newest observations for f = 0.10, 0.25, 0.50, 0.75 or 1.00, while keeping origins/targets fixed. Fitting still uses full training histories; reference capacity and cumulative exposure remain available.

Forecasting model equations

Six inputs per checkpoint; one GRU layer with eight hidden units; horizon H = 2; origin-relative output scale a = 0.02.

GRU encoder - same update convention as PyTorch

$$\mathbf{r}_k = \sigma(W_{ir}\widetilde{\mathbf{x}}_k + b_{ir} + W_{hr}\mathbf{h}_{k-1} + b_{hr})$$

$$\mathbf{z}_k = \sigma(W_{iz}\widetilde{\mathbf{x}}_k + b_{iz} + W_{hz}\mathbf{h}_{k-1} + b_{hz})$$

$$\mathbf{n}_k = \tanh\left(W_{in}\widetilde{\mathbf{x}}_k + b_{in} + \mathbf{r}_k \odot (W_{hn}\mathbf{h}_{k-1} + b_{hn})\right)$$

$$\mathbf{h}_k = (1 - \mathbf{z}_k) \odot \mathbf{n}_k + \mathbf{z}_k \odot \mathbf{h}_{k-1}$$

σ is the sigmoid, ⊙ is elementwise multiplication and the initial hidden state is zero. Packed sequences exclude padding from the final state. The representation is v = tanh(Wv Dropout(h_L) + bv); dropout probability is 0.1 during training and disabled at evaluation. Internal GRU dropout is zero because there is one recurrent layer.

Point head and ordered probabilistic head

$$\widehat{y}_{k+h} = y_k + a\,[W_o\mathbf{v} + b_o]_h$$

For the probabilistic model, a linear head produces three raw values $u_{h,0}$, $u_{h,1}$, $u_{h,2}$ per horizon:

$$q_{h,\,0.5} = y_k + a u_{h,\,0}$$

$$q_{h,\,0.1} = q_{h,\,0.5} - a\,\mathrm{softplus}(u_{h,\,1}), \quad q_{h,\,0.9} = q_{h,\,0.5} + a\,\mathrm{softplus}(u_{h,\,2})$$

softplus(u) = log(1 + exp(u)) makes the quantiles ordered. P10/P50/P90 refer to forecast quantiles, not battery health percentages. The network does not clip predictions to physical bounds; dispatch conversion validates whether a forecast defines a usable capacity.

Baselines use the same origins and labels

$$\boldsymbol{\phi} = \left[\widetilde{\mathbf{x}}_L,\ \frac{\widetilde{\mathbf{x}}_L - \widetilde{\mathbf{x}}_1}{L-1},\ L\right], \qquad \widehat{y}_{k+h} = y_k + g_h(\boldsymbol{\phi})$$

Ridge and XGBoost receive 13 inputs: six latest values, six per-checkpoint slopes and context length. For L = 1, slopes are zero. Persistence sets the residual g to zero. Ridge uses α = 1; XGBoost uses 150 trees, depth 2 and learning rate 0.03. These slopes are per checkpoint, not per day.

Training objective

The model learns its weights by minimizing a dimensionless loss; these training coefficients are separate from the dispatch wear weight λ.

Let b index training samples and h = 1, 2 index future checkpoints. y_{bh} is measured future SOH; $q_{bh,0.5}$ is the predicted median. All means below average the indicated sample/horizon entries; the quantile mean also averages three quantiles.

$$\mathcal{L}_{\text{point}} = \text{mean}_{b,h}\left[\left(\frac{q_{bh,0.5} - y_{bh}}{a}\right)^2\right], \qquad a = 0.02$$

$$\rho_\tau(e) = \max(\tau e, (\tau - 1)e), \qquad e_{bh\tau} = y_{bh} - q_{bh\tau}$$

$$\mathcal{L}_{\text{quantile}} = \frac{1}{a}\,\text{mean}_{b,h,\tau \in \{0.1, 0.5, 0.9\}}\rho_\tau(e_{bh\tau})$$

Soft physical preferences

Construct a checkpoint path $p_{b,0}$ = origin SOH, $p_{b,h}$ = predicted median, and increments $\Delta p_{b,h} = p_{b,h} - p_{b,h-1}$.

$$\mathcal{L}_{\text{mono}} = \text{mean}_{b,h}\left[\max\left(0, \frac{\Delta p_{bh} - \varepsilon}{a}\right)^2\right], \quad \varepsilon = 0.002$$

$$\mathcal{L}_{\text{smooth}} = \text{mean}_{b,h=2,\ldots,H}\left[\left(\frac{\Delta p_{bh} - \Delta p_{b,h-1}}{a}\right)^2\right]$$

The monotonic preference tolerates increases of 0.002 SOH, or 0.2 percentage points, before adding a penalty. Smoothness penalizes changes in adjacent checkpoint increments. Because calendar spacing varies, this is not a continuous-time degradation-rate law.

$$\mathcal{L}_{\text{PGP}} = \mathcal{L}_{\text{point}} + \mathcal{L}_{\text{quantile}} + 0.1\mathcal{L}_{\text{mono}} + 0.1\mathcal{L}_{\text{smooth}}$$

Model	Actual fitting objective
Vanilla GRU	Scaled point MSE only.
Quantile GRU	Scaled point MSE + pinball loss; both physical penalties set to zero.
Physics-guided GRU	The four-term PGP loss above.
Ridge / XGBoost	Residual regression; Ridge adds its L2 coefficient penalty. Persistence has no fit.

Adam: learning rate 0.003, weight decay 0.0001, gradient norm clipping at 1, maximum 200 full-batch epochs. Learning-rate reduction: factor 0.5 after 8 stale validation epochs. Early stopping: 25 stale epochs, minimum improvement 0.00001. The retained validation checkpoint excludes physical penalties from its validation score.

Evaluation and uncertainty

Leave-one-battery-out evaluation separates batteries, not randomly selected rows.

Each of six folds uses four training cells, one independent validation cell and one test cell. Validation is the next cell in sorted order after the test cell, wrapping around. The protocol repeats seeds 42, 43 and 44. Only training cells fit preprocessing and model parameters.

Point metrics

$$e_n = \hat{y}_n - y_n, \qquad \mathrm{MAE} = \frac{1}{N}\sum_n |e_n|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_n e_n^2}$$

$$\mathrm{MAPE} = \frac{100}{N}\sum_n \left|\frac{e_n}{y_n}\right|, \qquad R^2 = 1 - \frac{\sum_n e_n^2}{\sum_n (y_n - \bar{y})^2}$$

MAPE is omitted if any $|y| \le 10^{-8}$; R^2 is omitted for constant targets and can be negative otherwise. Multiply SOH MAE/RMSE by 100 to express percentage points. For capacity-Ah evaluation, multiply each target/prediction by that cell's reference capacity first.

Quantile metrics

$$\mathrm{Coverage}_{80} = \frac{1}{N}\sum_n \mathbf{1}\{q_{n,0.1} \le y_n \le q_{n,0.9}\}$$

$$\mathrm{Width} = \frac{1}{N}\sum_n (q_{n,0.9} - q_{n,0.1}), \qquad \hat{F}_\tau = \frac{1}{N}\sum_n \mathbf{1}\{y_n \le q_{n,\tau}\}$$

The nominal P10-P90 coverage target is 80%. Ideal empirical quantile fractions are 10%, 50% and 90%. Read coverage alongside interval width and mean pinball loss: a wider interval can improve coverage without improving precision.

Aggregation and current evidence

Compute metrics per cell/seed/horizon, average seeds within each cell, then average cells equally. Reported standard deviation is across these cell means. Horizon 0 in exports pools the one- and two-checkpoint prediction rows before computing each fold metric; it is not a zero-step forecast.

Model	MAE (SOH pp)	RMSE (SOH pp)	P10-P90 coverage
XGBoost	0.151	0.215	Not supplied
Ridge	0.199	0.267	Not supplied
Physics-guided GRU	0.200	0.274	86.4%
Quantile GRU	0.210	0.284	84.2%
Vanilla GRU	0.218	0.302	Not supplied
Persistence	0.378	0.439	Not supplied

XGBoost leads the aggregate reference run. Physical guidance does not establish superiority over all baselines. Thousands of prediction rows are repeated evaluations of six independent cells, not thousands of independent batteries.

Convert a forecast into wear

Forecast selection: latest held-out one-checkpoint PGP-GRU prediction per cell, seed 42, full history; selection does not use test error.

Convert relative SOH into nominal health

$$\gamma_i = \frac{Q_{i,0}}{Q_{\text{nom}}}, \qquad h_i = \gamma_i q_{i,\tau}, \qquad E_i^{\text{cap}} = E_{\text{nom}} h_i$$

Q_{nom} = 5 Ah and E_{nom} = 10 kWh. Use τ = 0.5 for median cases or τ = 0.1 for the conservative sensitivity. This is a cell-to-reference-pack scenario conversion, not a measured pack reconstruction.

Calibrate expected exposure without the held-out cell

$$\overline{F}_i = \frac{\text{median}\{\Delta A_{j,k} > 10^{-6} : j \in \mathcal{T}_i\}}{2Q_{\text{nom}}}$$

$$w_i = \max\left(0, \frac{(y_{i,\text{origin}} - q_{i,\tau})\gamma_i}{\overline{F}_i H_f}\right)$$

Training-cell positive increments in cumulative Ah estimate equivalent full cycles (EFC) per checkpoint. H_f is the selected forecast horizon, equal to 1 in dispatch. w is nominal capacity fraction lost per EFC. Predicted recovery produces w = 0 rather than negative wear. The validation and test cells are excluded from exposure calibration.

Price the linear wear proxy

$$F_i = \frac{\Delta t \sum_t (\eta c_t + d_t/\eta)}{2E_{\text{nom}}}, \qquad \Delta h_i = w_i F_i$$

$$C_{\text{wear}} = C_{\text{rep}} \frac{\Delta h_i}{B_{\text{loss}}}, \qquad C_{\text{rep}} = 100000 \text{ INR}, \quad B_{\text{loss}} = 0.20$$

c and d are external charging/discharging power in kW; η = √0.90 and Δt = 1 hour. EFC uses nominal pack energy, not health-adjusted capacity. The 0.20 capacity-loss budget is an economic assumption; the model does not enforce a measured end-of-life date.

G1 worked conversion	Value
Second-life reference / nominal cell capacity	4.5767 Ah / 5 Ah
Median forecast SOH	96.406% of second-life reference
Nominal health and available pack capacity	88.244% ; 8.824 kWh
Training exposure / checkpoint	8.403 EFC
Nominal capacity loss / EFC	0.0003671
Estimated wear price / EFC	₹183.56

This proxy omits temperature, calendar aging, SOC-dependent aging and validated pack-level behavior. It cannot establish lifetime extension or remaining useful life.

Storage optimisation model

A linear program over T = 720 hourly intervals. All INR costs and battery parameters are fixed inputs during a replay.

Given solar S_t and demand L_t, define direct solar $a_t = \min(S_t, L_t)$, surplus $s_t = S_t - a_t$ and unmet demand $\ell_t = L_t - a_t$. Decisions: charge c, discharge d, grid import g^+, grid export g^-, curtailment u (all nonnegative kW), and stored energy E (kWh).

Objective

$$\min J = \Delta t \sum_{t=0}^{T-1} \left(\pi_t^{\text{buy}} g_t^{+} - \pi_t^{\text{sell}} g_t^{-} \right) + \lambda C_{\text{wear}}$$

India C&I illustration: buy price ₹10/kWh flat; sell price ₹0/kWh; replacement assumption ₹1,00,000. These are not a DISCOM tariff or a supplier quote. The optional implemented carbon penalty is inactive because no carbon-intensity series is supplied.

Power balance and battery dynamics

$$c_t + g_t^{-} + u_t = s_t, \qquad g_t^{+} + d_t = \ell_t$$

$$E_{t+1} = E_t + \eta c_t \Delta t - \frac{d_t \Delta t}{\eta}, \qquad \eta = \sqrt{0.90}, \quad \Delta t = 1 \text{ h}$$

$$0 \le c_t \le P_{\max}, \quad 0 \le d_t \le P_{\max}, \quad P_{\max} = 3 \text{ kW}$$

$$0.10 E^{\text{cap}} \le E_t \le 0.90 E^{\text{cap}}, \qquad E_0 = E_T = 0.50 E^{\text{cap}}$$

The bounds apply to every stored-energy state. No-battery case: capacity and charge/discharge limits are zero. Full nominal case: capacity is 10 kWh. Second-life cases use forecast-adjusted capacity.

What these equations allow and prohibit

Solar serves demand first. Only solar surplus can charge storage, and discharge can only serve unmet demand. Thus grid charging, battery export and same-hour charging/discharging are excluded by the balance structure; no binary switching variables are needed. Exporting surplus remains physically allowed even though its revenue is zero.

The initial and final SOC equality is for the complete 720-hour horizon, not each day. It prevents apparent savings from consuming free initial stored energy without replacing it. The full solar/load horizon is known in advance; this is perfect-foresight replay, not a live predictive controller.

Solver: CVXPY with CLARABEL. Non-optimal results and constraint violations above 10^{-5} are rejected. Fixed health, constant linear wear and omitted demand charges mean this is an operating-scenario model, not a complete industrial project-finance model.

Policies and numerical review

The INR scenario changes operating economics. Historical EUR replay percentages must not be carried over without recalculation.

Policy	Capacity and wear basis	Decision objective
A / No battery	No usable storage	Grid energy cost after direct solar
B / Full nominal	10 kWh; same second-life wear coefficient for comparison	Energy cost only
C / Second-life	Median forecast capacity and wear	Energy cost only ($\lambda = 0$)
D / Wear-aware	Same capacity and wear as C	Energy cost + $\lambda \times$ wear; $\lambda = 0.5$, 1 or 2
P10 / Conservative	Lower health quantile; recalculated wear	Energy cost + wear ($\lambda = 1$)

Compare C against median D to isolate the effect of the operating policy. P10 changes assumptions as well as decisions. B does not model an independently calibrated new-battery wear law.

Current INR comparison: mean across six cell scenarios

Policy	Energy charge	Estimated wear	Combined	EFC
C	₹464.12	₹1,812.81	₹2,276.94	17.81
D / $\lambda = 1$	₹1,308.03	₹364.77	₹1,672.80	8.91

Reported combined cost always equals energy cost + full estimated wear. For $\lambda = 0.5$ or 2, this differs from the objective the solver minimizes. The table describes one proxy month, not observed monthly Indian site performance.

Why G1 can remain idle

For G1, C reduces the energy charge to ₹451.76 but incurs ₹3,292.67 of estimated wear. D at $\lambda = 1$ instead leaves the battery effectively idle: energy charge ₹2,153.50 and approximately zero wear. Under the specified proxy, cycling is not worth its estimated aging cost.

India C&I review boundary

The power input is the German OPSD residential profile, not an Indian industrial load. The 10 kWh / 3 kW pack is a reference system, not a sized factory installation. No demand-charge reduction, backup reliability, diesel displacement, subsidy, investment payback or field saving is established.

Before a commercial pilot: obtain site interval demand/solar, DISCOM tariff and ToD slots, sanctioned demand and power-factor terms, export agreement, pack quotation and measured aging. Add the relevant omitted costs and operational requirements before judging commercial viability.

Review checklist & traceability

Source audit performed against the local implementation and saved experiment artifacts on 26 October 2026.

Review question	Current status / decision needed
Are the six predictors available at inference?	Yes in the research table. Verify equivalent measurements and duty metadata at the target Indian site.
Is feature importance established?	Not by this attribute list. Static cell metadata may correlate with identity; test transferability and ablations separately.
Are physics penalties hard physical laws?	No. They are soft preferences; the neural forecast has no hard [0,1] clipping.
Are uncertainty and wear calibrated?	Coverage is evaluated on six cells; pack-level wear calibration remains outstanding.
Is this a complete C&I; economic model?	No. Demand/fixed charges, duties, taxes, installation, O&M;, financing, subsidies and backup value are excluded.
Has this document rerun training or modified the dashboard?	No. It formalizes the inspected implementation and reads the existing exported snapshot.

Implementation sources - SHA-256 prefixes in brackets

S1 / outputs/lobo/config.yaml [1b36a531b230]

S2 / src/second_life_battery/data/features.py [aa1a74c43c1e]

S3 / src/second_life_battery/data/preprocessing.py [0fd03062ad4b]

S4 / src/second_life_battery/data/sequences.py [72cccda68448]

S5 / src/second_life_battery/models/gru.py [78a7fb2c7fe8]

S6 / src/second_life_battery/models/physics_guided_gru.py [2aec0e2d7c1e]

S7 / src/second_life_battery/training/losses.py [b683aeae5784]

S8 / src/second_life_battery/training/trainer.py [4804f50d6026]

S9 / src/second_life_battery/optimization/degradation.py [f99a06eb54f6]

S10 / src/second_life_battery/optimization/solar_dispatch.py [4d819922edfb]

S11 / src/second_life_battery/evaluation/dashboard.py [f6df4f35ebae]

S12 / configs/india_ci.yaml [820e91cab354]

Data and results: data/processed/checkpoints.csv; outputs/lobo/summary.csv; outputs/lobo/predictions.csv; outputs/dispatch/wear_conversion.json; dashboard/dist/data.js. Primary dataset: Stanford Energy Control Lab, OSF 8JNR5, DOI 10.17605/OSF.IO/8JNR5. Solar/load: OPSD household data, version 2020-04-15, DE_KN_residential3. Dataset attributions are preserved; no new literature claim is made.